

EFC Closure Conjectures

Proposed Relations: $g^\dagger = cH_0/e$ and $C \approx 2e - 1$
A Testable Ansatz for EFC Transition Parameters

Morten Magnusson
Independent Researcher, Norway
ORCID: 0009-0002-4860-5095

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Abstract

We propose two closure relations for the Energy-Flow Cosmology (EFC) screening model: the transition acceleration $g^\dagger = cH_0/e$ and the galactic phase factor $C = 2e - 1$. These relations are motivated by the natural emergence of Euler's number e in the Core Lock exponential structure $f(S) = f_0 \exp(-S/\Delta S)$. The proposed $g^\dagger$ formula yields a Hubble constant prediction $H_0 = 70.2 \pm 16.8$ km/s/Mpc from SPARC data alone, consistent with all current measurements but not yet precise enough to discriminate between them. We present these as testable conjectures with explicit fail conditions, not as first-principles derivations.

Keywords: Energy-Flow Cosmology, radial acceleration relation, Hubble tension, transition acceleration, closure relations

Epistemic Status: Conjectured closure. Consistent with data but not derived from first principles. Testable.

1 Motivation

The EFC screening model for the Radial Acceleration Relation (RAR) takes the form:

$$\mu = \left(1 + \frac{g^\dagger}{g_{\text{bar}}}\right)^k \quad (1)$$

where $\mu = g_{\text{obs}}/g_{\text{bar}}$ is the gravitational enhancement factor, $k \approx 0.415$ is the screening exponent, and $g^\dagger \approx 2.5 \times 10^{-10}$ m/s² is the transition acceleration. The phase factor $C = k/a_G \approx 4.4$, where $a_G \approx 0.094$ is the universal coupling coefficient from H_0 tension analysis.

These parameters ($g^\dagger$, C , k) are currently fitted from SPARC data. A first-principles theory would derive them from more fundamental quantities. Here we propose candidate relations and assess their consistency.

Note: The transition acceleration $g^\dagger$ is defined within the EFC screening form $\ln \mu = k \ln(1 + g^\dagger/g_{\text{bar}})$. Numerical comparisons to MOND a_0 depend on the choice of parameterization; the two scales are related but not identical.

2 The Proposed Relations

2.1 Transition Acceleration: $g^\dagger = cH_0/e$

Dimensional analysis suggests that the transition acceleration should involve the cosmic expansion rate H_0 and the speed of light c :

$$g^\dagger = \frac{c \times H_0}{\eta} \quad (2)$$

where η is a dimensionless factor. We conjecture that $\eta = e$ (Euler’s number), motivated by the Core Lock structure:

$$f(S) = f_0 \exp\left(-\frac{S}{\Delta S}\right) \quad (3)$$

At $S = \Delta S$, the grid stiffness falls to f_0/e . We *define* the screening transition to occur when the stiffness has dropped by one e -fold from its reference value. If this threshold maps to the acceleration where screening transitions from inactive to active, we obtain:

$$\boxed{g^\dagger = \frac{cH_0}{e}} \quad (4)$$

This is a closure choice linking the cosmological scale cH_0 to the galactic transition via the natural exponential threshold.

2.2 Phase Factor: $C = 2e - 1$

The phase factor C describes how much more phase galactic structures accumulate compared to the cosmological background. We propose C as a compact closure constant built from the same exponential structure:

$$\boxed{C = 2e - 1 \approx 4.437} \quad (5)$$

The form $2e - 1$ is a minimal two-term expression matching the inferred $C \approx 4.4$ within current uncertainties. It can be written as $(e - 1) + e$, suggesting a structure where galactic accumulation combines the cosmological baseline phase $(e - 1)$ with one additional e -folding. However, this interpretation is heuristic; we have not derived C from an explicit integral over galactic structure.

2.3 Cosmological Phase

For $S_{\text{today}} \approx 1$ and $S_{\text{CMB}} \approx 0$, the cosmological phase difference is:

$$\Delta\Phi_{\text{cosmo}} \approx e^{S_{\text{today}}} - 1 \approx 1.7 \quad (6)$$

This is consistent with the Core Lock structure but depends on the assumed entropy mapping $S(z)$.

3 Numerical Consistency

Table 1: Comparison of conjectured relations and observed values

Parameter	Conjectured	Observed (SPARC)	Status
$g^\dagger$	cH_0/e (symbolic)	$(2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$	See note
C	$2e - 1 = 4.437$	4.4 ± 0.6	Consistent
$\Delta\Phi_{\text{cosmo}}$	$\approx e^{S_{\text{today}}} - 1$	~ 1.7	Consistent

Note on $g^\dagger$: The conjectured relation $g^\dagger = cH_0/e$ gives different values depending on H_0 :

- For $H_0 = 67.4 \text{ km/s/Mpc}$ (Planck): $g^\dagger = 2.41 \times 10^{-10} \text{ m/s}^2$
- For $H_0 = 73.0 \text{ km/s/Mpc}$ (SH0ES): $g^\dagger = 2.61 \times 10^{-10} \text{ m/s}^2$
- For $H_0 = 70.2 \text{ km/s/Mpc}$ (mean): $g^\dagger = 2.51 \times 10^{-10} \text{ m/s}^2$

The SPARC-measured value $(2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$ is consistent with all of these within its 24% uncertainty.

4 Non-Circular Test: H_0 from SPARC

To avoid circularity (using H_0 to predict H_0), we invert the relation:

$$H_0 = \frac{e \times g^\dagger}{c} \quad (7)$$

Using the SPARC-measured value $g^\dagger = (2.51 \pm 0.60) \times 10^{-10} \text{ m/s}^2$, we obtain:

$$H_0^{(\text{SPARC})} = 70.2 \pm 16.8 \text{ km/s/Mpc} \quad (8)$$

This prediction is consistent with Planck (67.4), SH0ES (73.0), and TRGB (69.8) measurements, well within 1σ in all cases. However, the large uncertainty (24%) prevents discrimination between competing H_0 values. The central value interestingly falls at the midpoint of the H_0 tension, but this may be coincidental given the uncertainties.

5 What Remains Free

Even with these conjectures, one parameter remains irreducible:

$$a_G \approx 0.094 \quad (9)$$

The coupling coefficient a_G sets the amplitude of the H_0 tension and cannot currently be derived from simpler quantities. All other parameters follow:

- $k = a_G \times C$
- $\Delta\Phi_{\text{cosmo}} \approx e^{S_{\text{today}}} - 1$ for $S_{\text{today}} \approx 1$
- H_0 offset via Friedmann: $a_{H_0} = \frac{1}{2}a_G$

6 Fail Conditions

These conjectures are falsified if:

1. **Improved $g^\dagger$ measurements** (with $<10\%$ uncertainty) are inconsistent with cH_0/e at $>3\sigma$, using an independently determined H_0 .
2. **The phase factor C** , when measured independently via hierarchical per-galaxy analysis, deviates from $2e - 1 = 4.437$ by more than 20%.
3. **Alternative functional forms** for $f(S)$ (e.g., power-law or Lorentzian) provide better fits to RAR data without natural factors of e .

7 Conclusion

We have proposed two closure relations for EFC: $g^\dagger = cH_0/e$ and $C = 2e - 1$. These are motivated by the exponential structure of Core Lock and are numerically consistent with SPARC data within current uncertainties. However, they remain conjectures until either (a) derived from explicit physical arguments, or (b) validated by improved measurements with smaller uncertainties.

The key testable prediction is:

$$H_0 = \frac{e \times g^\dagger}{c} \quad (10)$$

Current SPARC data yields $H_0 = 70.2 \pm 16.8 \text{ km/s/Mpc}$. Future surveys with better-constrained transition accelerations can test this relation precisely.

References

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Version History

Version	Date	Changes
1.0	February 2026	Initial specification